The PQC Hard Problem Colloquium

Lattice, Code, Hash, Multivariate, Isogeny, MPC/ZK

Instructor: Ji Yonghyeon

hacker3740@kookmin.ac.kr

Department of Cyber Security
Koomin Universeity

March 9, 2026

1. Introduction
2. Classical Foundations
3. Post-Quantum Hard Problem Families
4. The Colloquium
5. Preview

1 Introduction

2 Classical Foundations

3 Post-Quantum Hard Problem Families

4 The Colloquium

5 Preview

Purpose of This Colloquium

- This colloquium is about the **hard problems** that underlie modern public-key cryptography and post-quantum cryptography.
- The goal is not only to know *which schemes exist*, but to understand:
  - what mathematical problem each scheme relies on,
  - what the best known attacks are,
  - how classical and quantum complexity differ,

1 Introduction

2 Classical Foundations

3 Post-Quantum Hard Problem Families

4 The Colloquium

5 Preview

Classical Number-Theoretic Foundations

Historically, public-key cryptography was built on three dominant assumptions:

- **Integer Factorization** — RSA, Rabin
- **Discrete Logarithm in Finite Fields** — Diffie–Hellman, DSA, ElGamal
- **Elliptic Curve Discrete Logarithm** — ECDH, ECDSA, EdDSA

Integer Factorization

- Integer factorization underlies **RSA** and **Rabin**.
- Classically, the best known attacks include:
  - **Quadratic Sieve**,
  - **General Number Field Sieve**.
- In the quantum setting, **Shor's algorithm** solves the problem in polynomial time.

Discrete Logarithm and Elliptic Curve DLP

Finite-field DLP

- Supports Diffie–Hellman, DSA, and ElGamal.
- Best classical attacks include **Index Calculus** and **Number Field Sieve**.

Elliptic-curve DLP

- Supports ECDH, ECDSA, and EdDSA.
- Generic classical attacks are roughly $O(\sqrt{n})$, e.g. Pollard's rho.
- Yet both finite-field DLP and ECDLP fall to **Shor's algorithm** in polynomial time.

The Transition to PQC

Key Transition

The post-quantum transition is the replacement of Shor-vulnerable assumptions with alternative assumptions for which no efficient quantum algorithms are known.

- It is a transition in **mathematical foundations**.
- The main candidate families are:
  - lattices,
  - codes,
  - hash-based constructions,
  - multivariate systems,
  - isogenies.

1 Introduction

2 Classical Foundations

3 Post-Quantum Hard Problem Families

4 The Colloquium

5 Preview

Lattice Problems

The lattice family includes:

- **LWE**,
- **SIS**,
- **SVP**,
- **CVP**.

Representative systems include:

- Kyber / ML-KEM,
- Dilithium / ML-DSA,
- NTRU,
- fully homomorphic encryption constructions.

Code-Based Problems

Code-based cryptography is centered on:

- **syndrome decoding**,
- **quasi-cyclic syndrome decoding problem**
- more generally, decoding random linear codes.

Representative systems include:

- McEliece,
- HQC,
- BIKE.

Hash-Based Assumptions

Hash-based cryptography relies on:

- collision resistance,
- preimage resistance,
- second-preimage resistance.

Representative signature systems include:

- SPHINCS+,
- XMSS,
- LMS.

Multivariate Problems

Multivariate cryptography is based on the difficulty of solving systems of quadratic equations over finite fields.

This is commonly framed as the **MQ problem**.

Representative systems include:

- UOV,
- MAYO,
- historically, Rainbow.

Isogeny Problems

Isogeny-based cryptography is among the most mathematically elegant areas in PQC.

Representative names include:

- SQISign,
- CSIDH,
- historically, SIDH / SIKE.

1 Introduction

2 Classical Foundations

3 Post-Quantum Hard Problem Families

4 The Colloquium

5 Preview

Suggested Colloquium Flow

A natural progression for the colloquium is:

1. Classical number-theoretic assumptions and Shor's algorithm
2. Lattice problems and reduction-based security
3. Code-based assumptions and ISD
4. Hash-based signatures
5. Multivariate constructions
6. Isogeny cryptography
7. Cross-family comparison of security assumptions

Expected Outcomes

By the end of the colloquium, participants should be able to:

- explain the main hard-problem families in PQC,
- describe the best known attacks against each family.

Closing

- Post-quantum cryptography is not simply the search for new algorithms.
- It is the search for mathematical problems whose hardness survives quantum computation, structural cryptanalysis, and attack advances.

That is the central purpose of the PQC Hard Problem Colloquium.

1 Introduction

2 Classical Foundations

3 Post-Quantum Hard Problem Families

4 The Colloquium

5 Preview

A General Cryptographic Template

Let

- $\mathcal{I}$ is the public instance,
- $\mathcal{S}$ is the hidden witness, secret, or structure,

A hard problem is typically one of the following:

$$\text{Search: } \mathcal{A}(\mathcal{I}) = \mathcal{S}, \qquad \text{Decision: } \mathcal{A}(\mathcal{I}) \in \{0, 1\},$$

with no efficient known algorithm for the relevant parameter regime.

Integer Factorization

Instance. Let

$$N = pq,$$

where p, q are large primes. The public input is N.

Search problem. Recover nontrivial factors:

$$\text{Given } N, \text{ find } p, q \text{ such that } N = pq.$$

Discrete Logarithm Problem in Finite Fields

Let $G = \langle g \rangle$ be a cyclic group, typically

$$G = \mathbb{F}_p^{\times}.$$

Instance. Given

$$g \in G, \qquad h = g^x \in G.$$

Search problem. Recover the exponent x:

$$\text{find } x \text{ such that } h = g^x.$$

Elliptic Curve Discrete Logarithm Problem (ECDLP)

Let $E/\mathbb{F}_q$ be an elliptic curve, and let $P \in E(\mathbb{F}_q)$ be a point of large order n.

Instance. Given

$$P, \qquad Q = xP.$$

Search problem. Recover

$$x \in \mathbb{Z}_n \quad \text{such that} \quad Q = xP.$$

Lattice Problems

A lattice generated by basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m \in \mathbb{R}^n$ is

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \boldsymbol{b}_i : z_i \in \mathbb{Z} \right\}.$$

Shortest Vector Problem (SVP). Find

$$\boldsymbol{v} \in \Lambda \setminus \{\boldsymbol{0}\}$$

minimizing

$$\|\boldsymbol{v}\|.$$

Closest Vector Problem (CVP). Given $\boldsymbol{t} \in \mathbb{R}^n$, find

$$\boldsymbol{v} \in \Lambda$$

minimizing

$$\|\boldsymbol{t} - \boldsymbol{v}\|.$$

Learning With Errors (LWE)

Secret. $\boldsymbol{s} \in \mathbb{Z}_q^n$.

Samples. For random $\boldsymbol{a}_i \in \mathbb{Z}_q^n$, we observe

$$b_i = \langle \boldsymbol{a}_i, \boldsymbol{s} \rangle + e_i \pmod{q},$$

where e_i is a small error.

Search-LWE. Given many pairs

$$(\boldsymbol{a}_i, b_i),$$

recover $\boldsymbol{s}$.

Decision-LWE. Distinguish

$$(\boldsymbol{a}_i, \langle \boldsymbol{a}_i, \boldsymbol{s} \rangle + e_i)$$

from uniform samples in $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

SIS: Short Integer Solution

Let

$$A \in \mathbb{Z}_q^{m \times n}.$$

Problem. Find a nonzero vector

$$\boldsymbol{x} \in \mathbb{Z}^n$$

such that $A\boldsymbol{x} \equiv \mathbf{0} \pmod{q}$ and $\|\boldsymbol{x}\|$ is small.

Code-Based Hardness: Syndrome Decoding

Let

$$H \in \mathbb{F}_2^{r \times n}$$

be a parity-check matrix.

Given. A syndrome

$$\boldsymbol{s} \in \mathbb{F}_2^r$$

and a weight bound w.

Problem. Find an error vector

$$\boldsymbol{e} \in \mathbb{F}_2^n$$

such that $H\boldsymbol{e}^T = \boldsymbol{s}^T$ and

$$\mathrm{wt}(\boldsymbol{e}) \leq w.$$

Hash-Based Hardness

Let

$$H : \{0,1\}^* \to \{0,1\}^n$$

be a cryptographic hash function.

Preimage resistance. Given $y \in \{0,1\}^n$, find

$$x \text{ such that } H(x) = y.$$

Second-preimage resistance. Given x, find

$$x' \neq x \quad \text{such that} \quad H(x') = H(x).$$

Collision resistance. Find

$$x \neq x' \quad \text{such that} \quad H(x) = H(x').$$

Thank you for listening !

Ji, YongHyeon

hacker3740@kookmin.ac.kr